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Palaeomagnetism of Permo-Triassic rocks in the Iberian Cordillera, Spain: acquisition of secondary and characteristic remanence

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Abstract: Magnetostratigraphy is widely used as a tool for stratigraphical correlation but there are problems in its application to rocks which have suffered complex diagenesis, particularly sandy, continental red beds. In such cases the complete magnetic record should be studied in relation to its depositional, diagenetic and structural history. To this end a detailed palaeomagnetic study has been made of Permian and Triassic rocks in the northern Iberian Cordillera, Spain. Four stratigraphical units comprising the Autunian, 'Saxonian', Buntsandstein and Muschelkalk have been studied in a single section near Molina de Aragon which is well-documented sedimentologically. Autunian rocks have a characteristic remanence which is exclusively reversed and corresponds to the Kiaman Interval of Lower Permian age. A pole position (49 N, 167 W) is consistent with previous studies which indicate the rotation of Iberia in post-Triassic times relative to the stable European plate. The Saxonian facies show complex, multicomponent magnetizations; no polarity zonation can be resolved but individual components were acquired prior to the rotation of Iberia.

The Buntsandstein is remarkable in that much of it is completely remagnetized in a direction similar to the present local geomagnetic direction. This secondary remanence is carried by fine grained hematite which has been formed as a result of carbonate dissolution and other diagenetic reactions. Initially, these iron oxides were probably superparamagnetic and remanence was only blocked-in by structural inversion of the Iberian Cordillera in Tertiary times and after rotation of the Iberian plate. Some of the primary remanence is preserved in some fine grained lithologies of the Buntsandstein, indicating that normal and reversed zones of magnetization were originally present. The magnetization of the Muschelkalk carbonates is also complex; secondary components similar to those of the Buntsandstein are present but there is evidence that primary components were mainly normal during Karnian times.

Our knowledge of the changes in the polarity of the Earth's magnetic field are detailed for most of the Mesozoic (Jurassic–Cretaceous) and Cenozoic partly because of palaeomagnetic studies of sedimentary sequences but principally because of the record of marine magnetic anomalies (Berggren *et al.* 1985; Kent & Gradstein 1985). Extending the polarity-reversal sequence back beyond the Jurassic has not been achieved with general consensus, although local successions have been studied in detail (Burek 1968; McIntosh *et al.* 1985).

In their pioneering work in the southern hemisphere Irving & Parry (1963) established a long period of reversed polarity (The Kiaman Magnetic Epoch) which was subsequently demonstrated to be present on a world-wide basis (Khramov 1971; McElhinny 1969). The fact that such a global correlation is possible demonstrates that magnetic stratigraphy can be a valuable tool in stratigraphic correlation. Extending the known polarity reversal scale backwards in time is of considerable importance but problems associated with remagnetization must be adequately resolved.

The conditions under which the ambient magnetic field is most efficiently recorded in a sedimentary sequence are:

(1) A reversal rate of the Earth's magnetic field which is not too rapid. Frequent reversals relative to the sedimentation rate are likely to result in the acquisition of composite magnetizations and the obfuscation of the magnetic record.
(2) A sedimentation rate which is neither too fast nor too slow. Fast sedimentation rates may result in the dilution of magnetic minerals in the sediment with only weak magnetic signals recorded. Too slow sedimentation rates may result in the contraction of the magnetic reversal stratigraphy to such an extent that successive periods of normal and reversed magnetization may be superimposed.
(3) Absence of substantial diagenetic changes. Diagenesis may alter the original magnetic record by oxidation of the detrital magnetic grains and the precipitation of authigenic magnetic phases. The development of chemical remanent remagnetization (CRM) in this way is a particular problem in red beds and why their value in magnetostratigraphic studies is limited. This problem is discussed in detail in the later part of this paper.

There have been a number of studies of the magnetic stratigraphy of the Permian and Triassic Systems but in NW Europe and N America interpretation is complicated by the fact that these Systems comprise very largely of continental red beds and carbonate–evaporite sequences which are

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strongly affected by diagenesis. As a result there has been much controversy as to the validity of published Permo-Triassic polarity-reversal sequences (Helsley 1969; Elston & Purucker 1979; Walker *et al.* 1981), even though reversal sequences are known in the Triassic and can be correlated over some distance (Lienart & Helsley 1980). For example, Herrero-Bervera & Helsley (1983) described an apparent polarity transition in a 1 m interval between normally (below) and reversely magnetized rocks (above) and Larson & Walker (1985) subsequently contended that the intermediate directions in this zone were produced by the superimposition of antiparallel components at least one of which was produced during diagenesis. It seems to us that the interpretation of magnetic polarity sequences (and transitions) in continental red beds will remain equivocal until there is adequate geochronological control which enables correlation with well-calibrated time equivalent sequences.

In this paper our aim is to discuss the origin of the magnetization and determine if secondary (diagenetic) remanence can be isolated to reveal characteristic magnetizations. There is evidence that some lithologies, mainly fine-grained siltstones carry some characteristic magnetization and some contribution to magnetostratigraphy can be made. In the Spanish Permo-Triassic rocks studied in this paper there is quite unequivocal evidence that much of the magnetic remanence originated during diagenesis. This is in the form of palaeomagnetic directions which lie near the present geomagnetic field and also diagenetic studies which show abundant authigenic hematite as pore-lining and pore-filling phases. In particular we can position the upper boundary of the Kiaman Epoch with reference to the biostratigraphy and we discuss the origin and significance of both the characteristic and secondary magnetizations.

Stratigraphical and sedimentological background

At the end of the Hercynian orogeny (Late Carboniferous–Early Permian), the Iberian Peninsula suffered a complex wrench faulting process with two main east–west systems (the northern Pyrenean–Bay of Biscay system and the Gibraltar system). An important NW–SE and NE–SW fault system developed in the interior of the Iberian Plate, forming different basins where continental sediments and associated volcanic rocks accumulated (Autunian). Tectonic activity decreased during the Permian with the deposition of Saxonian facies.

During Late Permian the older wrench fault-systems were first reactivated as normal faults which resulted in a series of basins (Sopeña *et al.* 1988). The basins show a comparable evolution: opening with continental sedimentation (Buntsandstein facies) evolving to shallow marine carbonates (Muschelkalk facies) followed by evaporites and red clays (Keuper facies), the classic Germanic facies, but clearly heterochronous in the different basins.

The area studied lies in the western part of the Iberian Cordillera in Central Spain (Fig. 1) and comprises rocks ranging in age from Early Permian to Late Triassic. The complete section has been studied in the Sierra de Aragoncillo (260 m thick) and the irregular distribution is due not only to the original palaeotopography but also it is related to the distribution of the volcanic rocks. The Lower Permian (Autunian) rests always unconformably on Hercynian basement and is covered also unconformably by Saxonian or Buntsandstein red beds. Sedimentation began with a volcaniclastic sequence of tuffs, ash flows and ignimbrites of lahar type (Muñoz *et al.* 1985). The sequence is calc-alkaline in character. These sedimentation events were associated with high energy sheet floods or locally channelled sands with abundant plant remains. These sediments evolve up into thin-bedded siltstones, and dark limestones that were deposited in a lacustrine environment. The Autunian age of the 'Capas de la Ermita' has been well characterized palynologically (Ramos *et al.* 1976).

The Saalian movements and subsequent erosion modified the regional palaeotopography greatly. Continental red beds ('Capas de Montesoro' of the Saxonian facies) consisting mainly of red mudstones with intercalated breccias of quartzite, sandstones, conglomerates, and thin carbonate beds of mainly alluvial origin, indicate a climatic change to more arid conditions. A number of small alluvial fans came into being in which the distinction between proximal and distal facies is hardly possible. There is no available biostratigraphical data for the Saxonian facies, which may reach a thickness of approximately 45 m in this area.

Buntsandstein sediments unconformably overlie all previous rocks and began with a period of extensive continental sedimentation on older Palaeozoic massifs in the Iberian Peninsula. Traditionally the basal unconformity has been considered to be the Palaeozoic–Mesozoic boundary, but the age of the lowest sediments is variable, ranging from Late Permian to Middle or Late Triassic. A Thuringian microflora has been recovered in our area from the lowest Buntsandstein unit (Conglomerados de la Hoz del Gallo) (Ramos & Doubinger 1979). Tectonics and sedimentation were clearly related; the basin was affected by a series of horst and graben with some areas of major subsidence, forming internal unconformities in the border of the basin, (Sopeña *et al.* 1983). The area studied is a major zone of subsidence, where Buntsandstein facies exhibit a thick development with up to 550 m present. Buntsandstein facies have been subdivided into six units (Ramos 1979) (Fig. 2) ranging in age from Thuringian to Ladinian. Buntsandstein facies mainly consist of conglomerates and sandstones red beds. The lower siliciclastic complex 'Conglomerados de la Hoz del Gallo', 'Areniscas de Rillo de Gallo', 'Nivel de Prados' and 'Areniscas del Rio Arandilla') is closely related to tectonic events (Ramos *et al.* 1986, Sopeña *et al.* 1988), noteably an alluvial complex related to late Hercynian faults developing during the sedimentation. The 'Conglomerados de la Hoz del Gallo' onlap 'Capas de Montesoro', the result of fan-lobe progradation produced by low-sinuosity channels (Ramos & Sopeña 1983) with a NNE trend. The overlying units ('Areniscas de Rillo de Gallo', 'Nivel de Prados' and 'Areniscas del Arandilla') accumulated as the result of braided fluvial systems producing transverse or linguoid bars that change upwards into channel fill deposits (Ramos *et al.* 1976). The 'Nivel de Prados' were deposited by higher-sinuosity streams with a new reactivation for the 'Areniscas del Rio Arandilla' (lower sinuoisty, coarse-grained) related to tectonic movements. The 'Lutitas y Areniscas de Rillo' unit display evidence of evolution from a distal fluvial environment to one of tidal influence, that probably took place during the late Anisian as indicated by a palynological assemblage in the upper part (Ramos 1979). This marks the beginning of the Tethyan transgression over this area of

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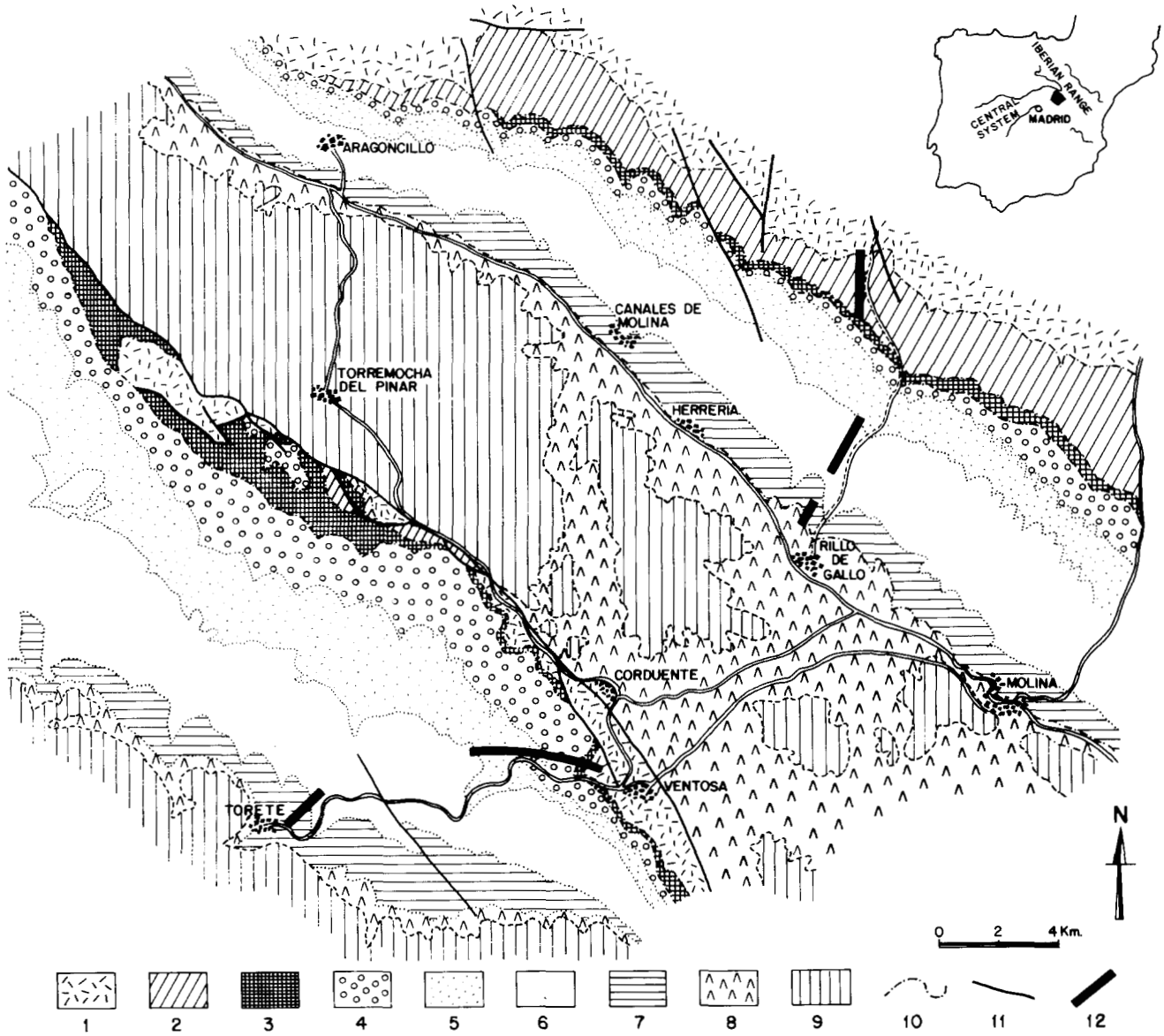


Fig. 1. Geological sketch and location map of the region. (1) Lower Palaeozoic. (2) Autunian: 'Capas de la Ermita'. (3) Saxonian: 'Capas de Montesoro'. (4) Buntsandstein: 'Conglomerados de la Hoz del Gallo'. (5) 'Areniscas de Rillo de Gallo' and 'Nivel de Prados'. (6) 'Areniscas del Rio Arandilla', 'Areniscas de Rillo' and 'Limos y Areniscas abigarrados de Torete'. (7) Muschelkalk: 'Dolomias de Tramacastilla' and 'Dolomias, Margas y Calizas de Royuela'. (8) Keuper. (9) Jurassic. (10) Unconformity. (11) Main Faults. (12) Studied sections.

western Europe. The 'Lutitas y Areniscas abigarrados de Torete' unit, with several palynological assemblages of Ladinian age, shows clear evidence of a supratidal environment in a semiarid climate with algal lamination and evaporite precipitation. The Muschelkalk carbonate facies represent the development of a more homogenous marine environment, with more general subsidence and widening of the basin. These last events took place in this area during Late Ladinian–Early Karnian as proved by palynological assemblages (Ramos 1979).

Palaeomagnetic sampling and measurement

Over 200 hand samples were collected in Permo-Triassic rocks of the Iberian Cordillera from the sections described above. Of these, 104 were from sections at Rio de Gallo and Hoz del Gallo near Molina de Aragon in Central Spain (Fig. 1). All the samples were orientated using both suncompass and magnetic compass. The exact stratigraphic horizon from which the samples were taken was noted using detailed sedimentological logs drawn up by Ramos (1979). A minimum of six 2.2 cm long by 2.5 cm diameter cores (specimens) were taken from each sample for palaeomagnetic studies. Thin and polished sections were made of the great majority of samples in order to identify any diagenetic changes and the oxide mineralogy. XRD and SEM studies were also carried out on the majority of samples.

The combination of all these techniques enables a comprehensive study of the relationships between palaeomagnetism and diagenesis in these Spanish Permo-Triassic red beds to be carried out.

At least one specimen from each of the hand samples was subjected to partial thermal demagnetization and a number of selected specimens were also analyzed using

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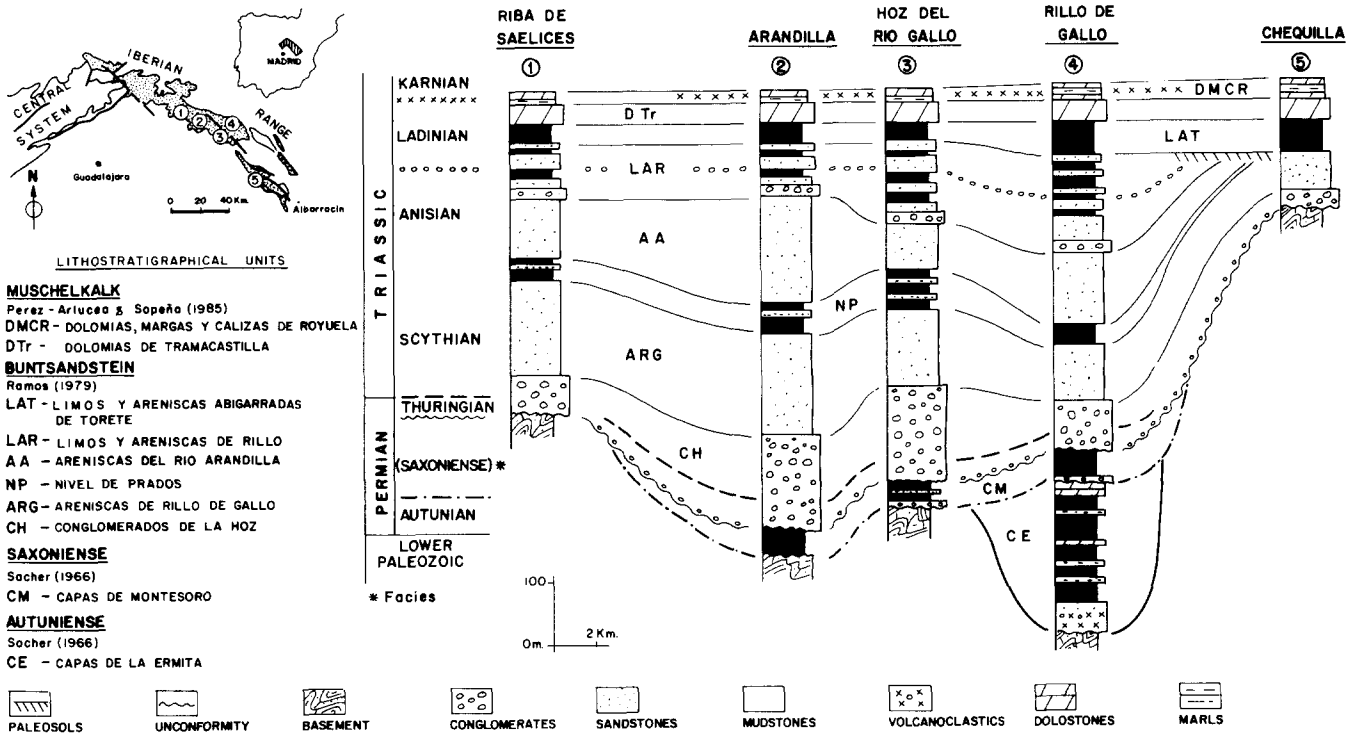


Fig. 2. Representative vertical profiles and generalized Permian–Triassic stratigraphy.

alternating field demagnetization (AF) methods. The measurements were made mainly at the Nuffield palaeomagnetic laboratory, Newcastle University and partly in the Department of Geological Sciences at Aston University. Thermal demagnetization was performed using the stepwise method and remanence measurements made with a two-axis cryogenic and Digico Spinner magnetometers. The selection of temperature increments was customized for each of the main stratigraphical units because of variations in the nature of the characteristic remanence (blocking temperature spectra) and after each heating step and remanence measurement the low field initial susceptibility was measured to monitor chemical changes. The various stratigraphical units are described below. In no case were temperature increments of more than 100 °C used and in many cases increments of only 30–50 °C were used above 500 °C.

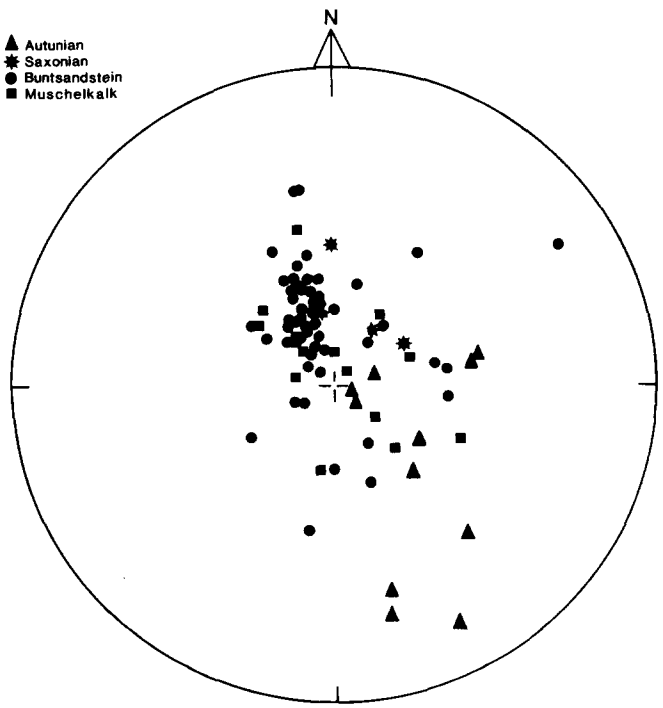


Fig. 3. Initial mean NRM directions of Permo-Triassic rocks from the area near Molina de Aragon, Iberian Cordillera, Spain.

All the data were treated using a variety of statistical techniques. In particular, individual components of magnetization have been isolated using the techniques described by Dunlop (1979) and Kirschvink (1980).

The initial NRM measurements show considerable variation (Fig. 3). Specimens from the Buntsandstein are, in general, tightly grouped near to the local geomagnetic direction (Dec: 354 Inc: +57°). The Autunian, however, show shallower inclinations and are strung out in SSE directions toward the known Permian direction for Central Spain (Van der Voo 1967). It should be noted that this direction differs from that of stable Europe because of the rotation of Iberia during Cretaceous times.

Autunian Facies

Fifteen horizons were sampled in the Autunian section of the Rio de Gallo including representatives of most of the facies. The initial NRM intensity varied between these facies from extremely weak organic-rich carbonates ($<0.1\ mA\ m^{-1}$) to slightly stronger volcaniclastics ($1.0\ mA\ m^{-1}$) with an average of ($0.64\ mA\ m^{-1}$). All were subjected to thermal demagnetization at 100 °C intervals

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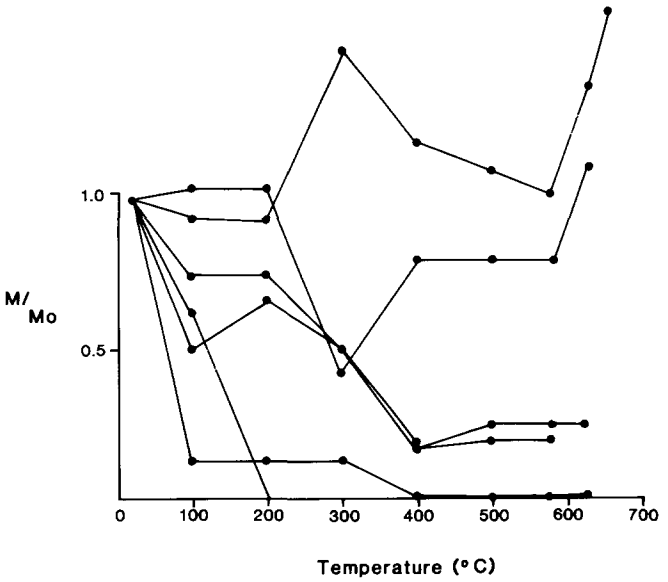


Fig. 4. Normalized intensity decay curves for Autunian rock.

during which there was rapid loss of NRM with a wide range of blocking temperature spectra (Fig. 4). There is also considerable variation in initial susceptibility with respect to temperature with large increases above 300–400 °C resulting from chemical changes (principally oxidation of fine-grained sulphide minerals).

The directional behaviour of a typical specimen (RG 6.3.1) is shown in Fig. 5. A steep positive component of magnetization is removed up to 200 °C and above this temperature a stable component is isolated with SSE declination and shallow inclination. The mean direction of fifteen components which have been isolated from five sample horizons in the Autunian section is Dec: 172°, Inc: −1°, α_{95}: 9.3°. The corresponding pole position is 49N167W. This is typical of other Permian results in the Iberian Peninsula (Van der Voo 1969) and differs from the Permian of stable Europe because of the rotation of Iberia in Cretaceous times.

Saxonian Facies

Only a small number (5) of horizons could be sampled in the Saxonian section of the Rio de Gallo, because of the lithological variation with abundant conglomerates, siltstones and rare sandstones. The initial NRM intensities are higher than all other lithologies in the Permo-Triassic section (average 6.81 mA m^{-1}).

During thermal demagnetization all the Saxonian specimens show complex behaviour. Specimen MA 115.4 is shown as a typical example (Fig. 6). There are three components which are most clearly seen on the stereographic projection (Fig. 6A). A component (A) with low

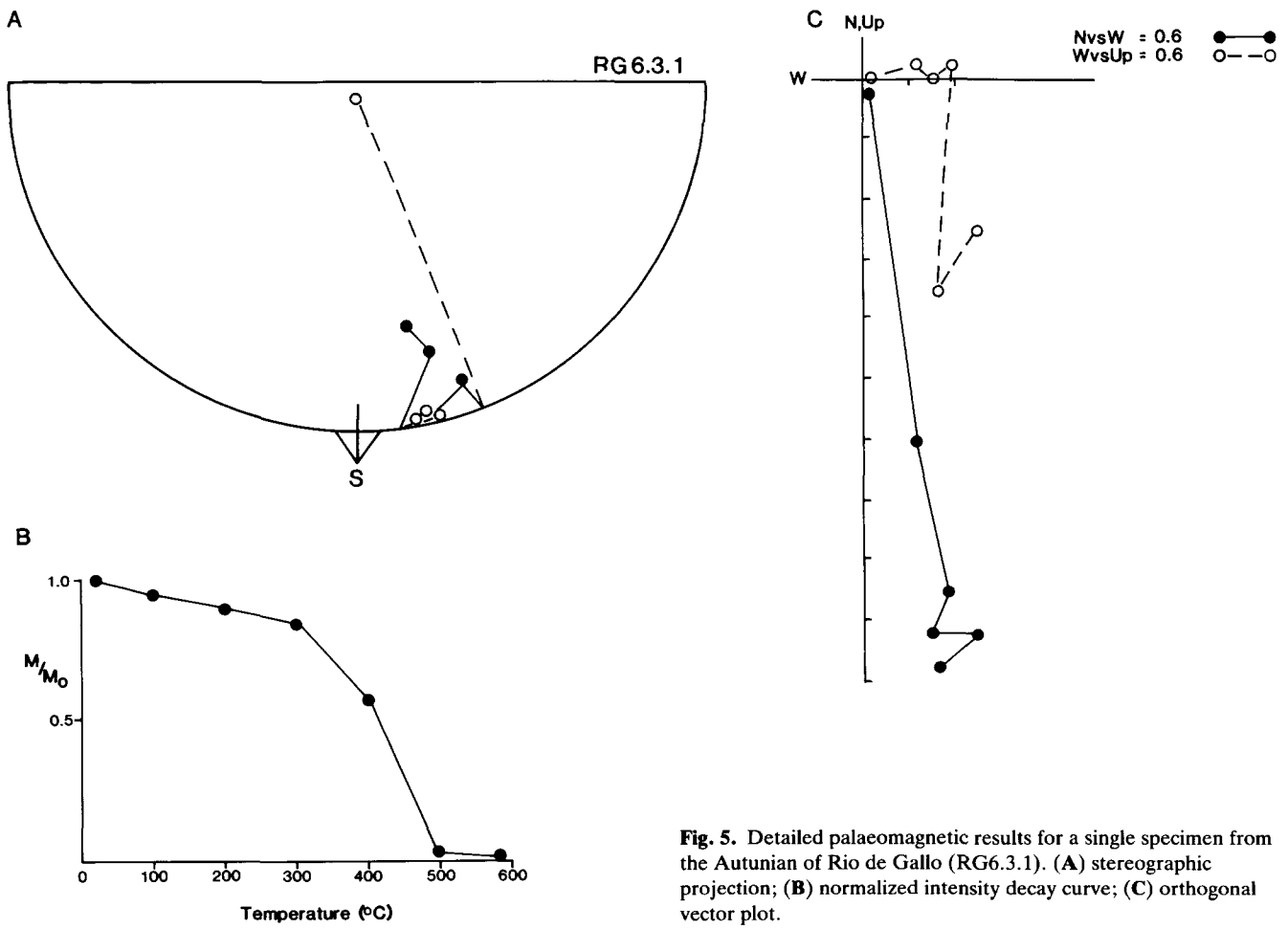


Fig. 5. Detailed palaeomagnetic results for a single specimen from the Autunian of Rio de Gallo (RG6.3.1). (**A**) stereographic projection; (**B**) normalized intensity decay curve; (**C**) orthogonal vector plot.

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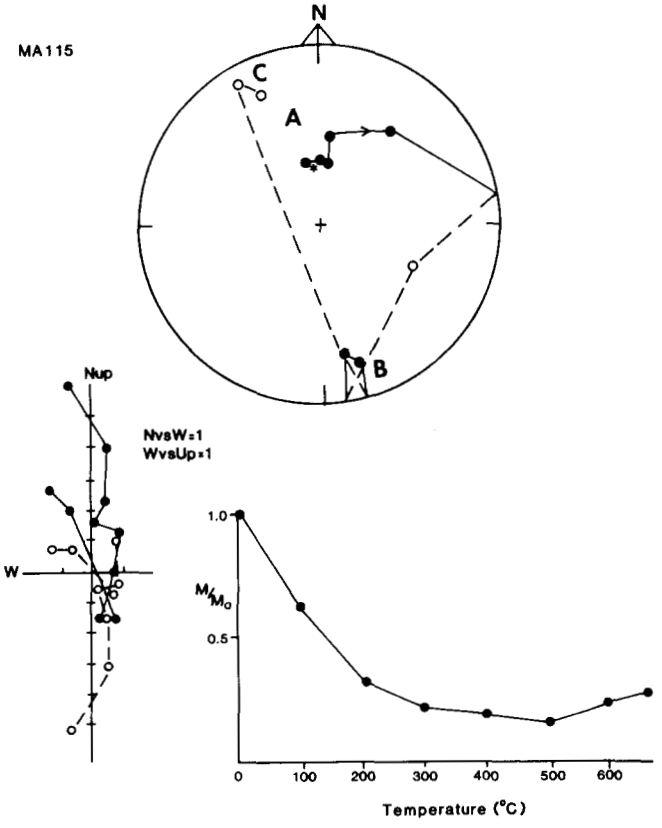


Fig. 6. Detailed palaeomagnetic results for a single specimen from the Saxonian of Hoz del Gallo (MA115.4). (**A**) stereographic projection; (**B**) normalized intensity decay curve; (**C**) orthogonal vector plot. The asterisk shows the present-day geomagnetic direction.

blocking temperatures (0–300 °C) lies very near to the local geomagnetic direction. The higher temperature components B and C) are almost perfectly antiparallel with the B-component of Dec: 160, Inc: +25 and Dec: 331, Inc: −12 being very close to the reversed Permian directions of the Autunian. On this basis the C-component is interpreted as a normal component.

For the purposes of establishing the magnetic stratigraphy of the Permo-Triassic the Saxonian facies is interpreted as a mixed polarity interval. No discrete zones of normal or reversely magnetized rock could be identified. It is not clear whether the multicomponent magnetizations are due to rapid field reversals or diagenetic overprinting over a long time during the Saxonian Interval. The fact that isolated components lie along a NNW–SSE axis is taken as a possible inclination that these components were acquired prior to the rotation of Iberia in Cretaceous times (Permo-Triassic directions from the stable European plate lies on a NE–SW axis).

Buntsandstein Facies

The Buntsandstein is the thickest and most variable unit in the Permo-Triassic of the Iberian Cordillera. A total of 60 samples were collected with representatives from all the stratigraphical units present (Fig. 2). The initial NRM measurements, with only a few exceptions are tightly grouped near to the local geomagnetic direction (Fig. 3). The initial intensities average 4.80 mA m^{-1}.

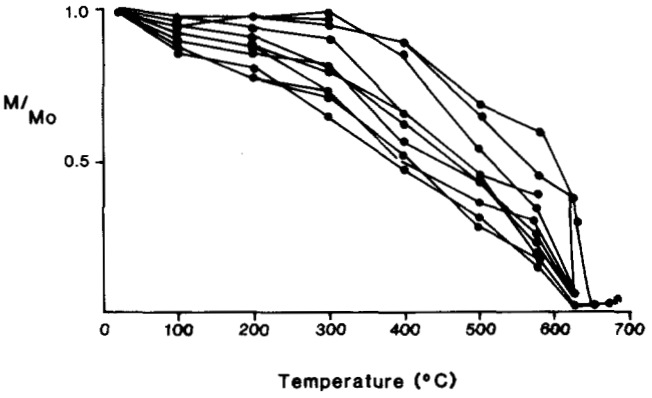


Fig. 7. Representative normalized intensity decay curves for Buntsandstein specimen.

At least one specimen from each of the samples was subjected to partial thermal demagnetization. During this procedure a wide variety of behaviour was revealed which contrasts with that seen in the underlying Autunian and Saxonian Facies. Despite this the blocking temperature curves show remarkable uniformity with an absence of substantial low blocking temperatures with about 50% of NRM removed by 500 °C and an important maximum blocking temperature at 630 °C (Fig. 7). On the basis of the directional and normalized intensity changes the following types of magnetization are distinguished.

Specimens containing two stable components of magnetization:

(A) A lower blocking temperature component lies near the local geomagnetic field direction and is superimposed on a higher blocking temperature with NNW or SSE declination and shallow inclinations. The blocking temperature window separating the two components is usually between 300° and 630 °C.

(B) Magnetizations comprising a stable component (0°–630 °C) in the local geomagnetic direction superimposed on a weak, shallow component which cannot be precisely defined.

Specimens containing one stable component of magnetization:

(C) Magnetizations comprising only a steep component near the local geomagnetic direction and in which no remnant of original magnetization can be identified.

Type A

Some 18% of the Buntsandstein specimens have magnetizations of this type. They occur in the (finer grained) upper part of the Limos y Areniscas de Rillo (Fig. 2). Examples of the Type A magnetization (MA 77.1A and MA 88.3.1U) are shown in Figs 8 and 9 respectively. In MA 77.1A the lower blocking temperature (secondary) component is relatively shallow with Dec: 352, Inc: +28 after bedding correction. Above 600 °C this component is removed and a higher blocking temperature component with more westerly direction and negative inclinations is isolated. In MA 88.3.1 the lower blocking temperature component is a steep vertical component with NNW declination. The orientation of the component can only be deduced from the orthogonal vector plot (Fig. 9). Note that its removal is associated with a blocking temperature at 100 °C followed by a slight, but steady rise in the normalized intensity decay. At 400 °C (the

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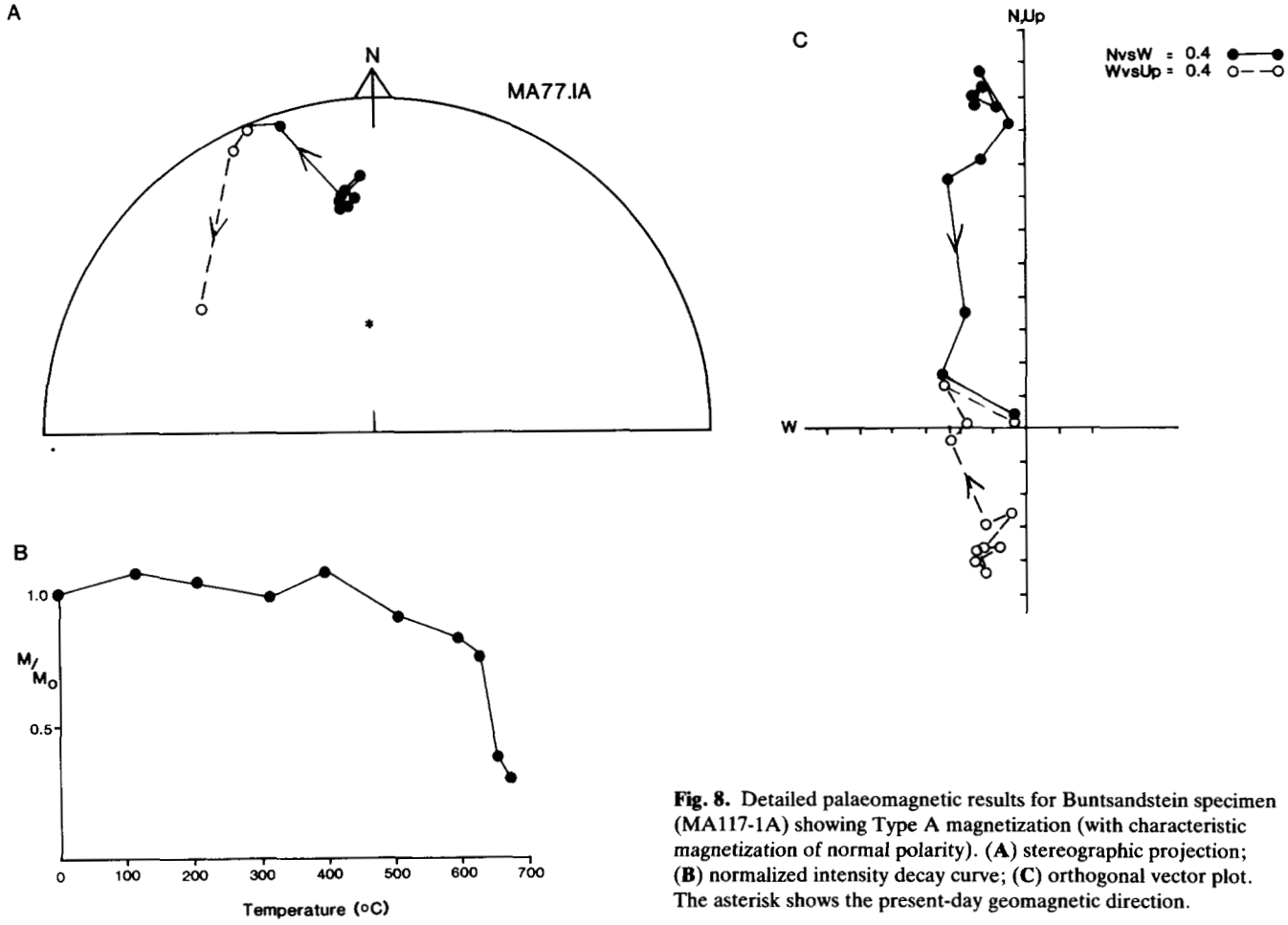


Fig. 8. Detailed palaeomagnetic results for Buntsandstein specimen (MA117-1A) showing Type A magnetization (with characteristic magnetization of normal polarity). (**A**) stereographic projection; (**B**) normalized intensity decay curve; (**C**) orthogonal vector plot. The asterisk shows the present-day geomagnetic direction.

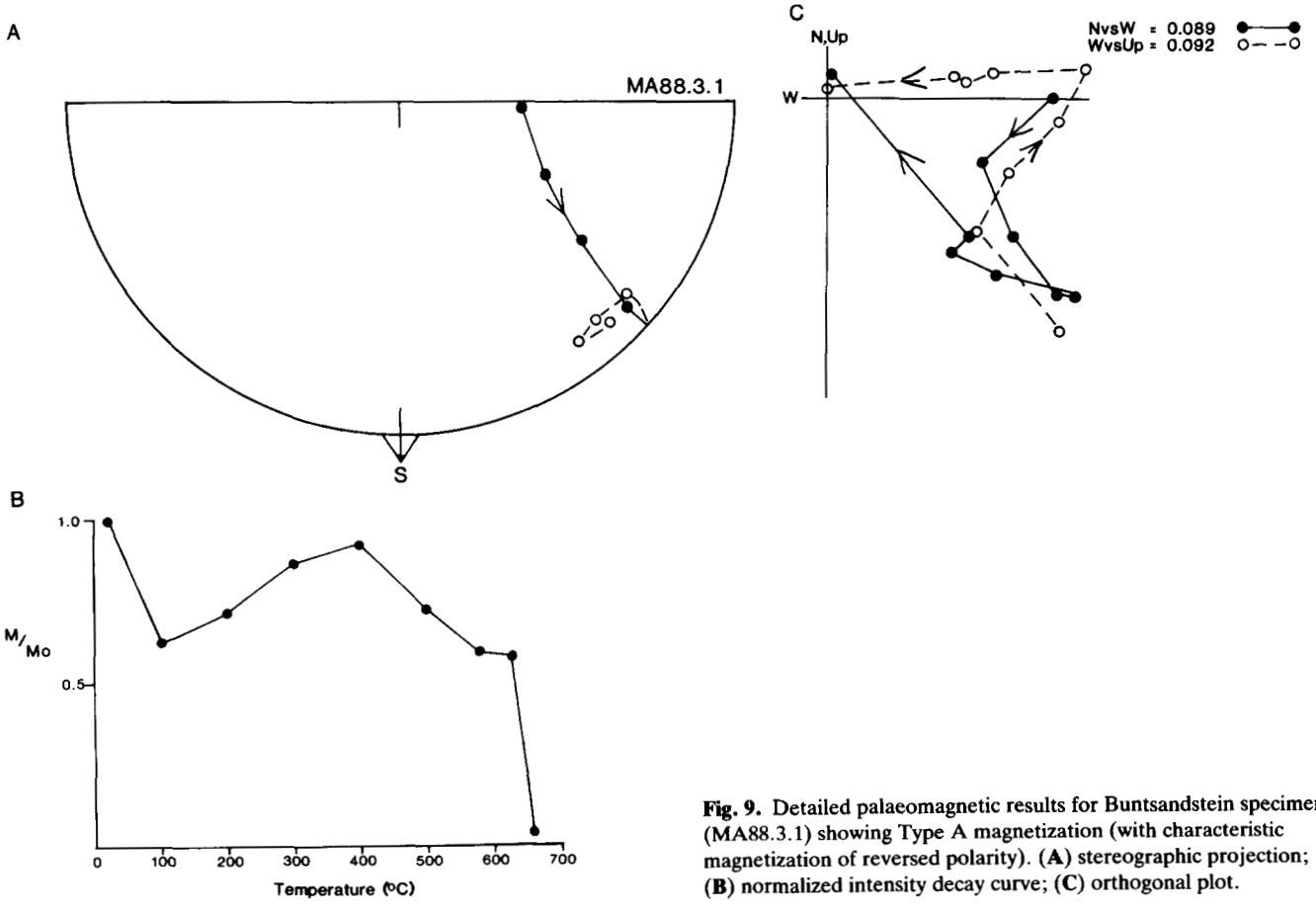


Fig. 9. Detailed palaeomagnetic results for Buntsandstein specimen (MA88.3.1) showing Type A magnetization (with characteristic magnetization of reversed polarity). (**A**) stereographic projection; (**B**) normalized intensity decay curve; (**C**) orthogonal plot.

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peak on the normalized intensity curve) the directional movement ceases and there is little or no change up to 660 °C. This stable component has Dec: 136.5; Inc: −4. It is noteworthy that this component is nearly antiparallel to the component isolated in MA 77.1A and on this basis the former is interpreted as a Permo-Triassic normal and the latter a Permo-Triassic reversed component. The origin of the secondary magnetization will be discussed after the description of the Bunstandstein B and C types of magnetization.

Type B

The majority (70%) of the Buntsandstein specimens have magnetizations of this type. They consist almost entirely of a single component of magnetization, lying near to the present Earth field direction for the area, and which is very stable during partial thermal demagnetization. A typical example is shown in Fig. 10 (MA 49.4.1). The initial direction of Dec: 338, Inc: +61 is virtually unchanged up to 600 °C. There is only a slight loss of NRM up to 300 °C with no associated directional change. More rapid loss above this temperature is not associated with directional change until 580 °C. Above this temperature the direction moves towards a shallower inclination in the SE quadrant. At 630 °C only 3% of the original remanence (1.42 mA m^{-1}) remains and reliable directions cannot be measured above this temperature. The clear movement of the direction close to a position interpreted as reversed Permo-Triassic suggests that

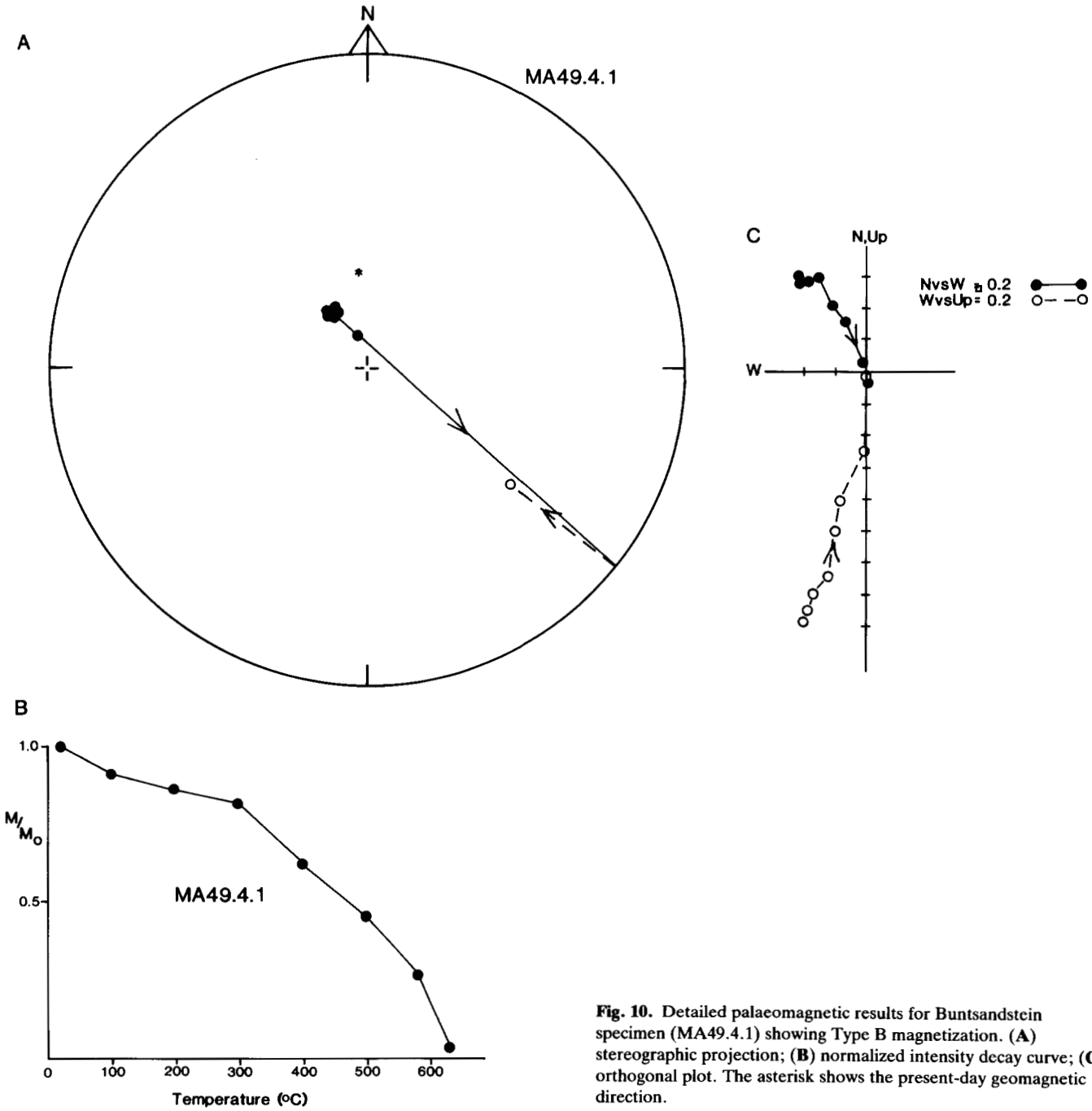


Fig. 10. Detailed palaeomagnetic results for Buntsandstein specimen (MA49.4.1) showing Type B magnetization. (**A**) stereographic projection; (**B**) normalized intensity decay curve; (**C**) orthogonal plot. The asterisk shows the present-day geomagnetic direction.

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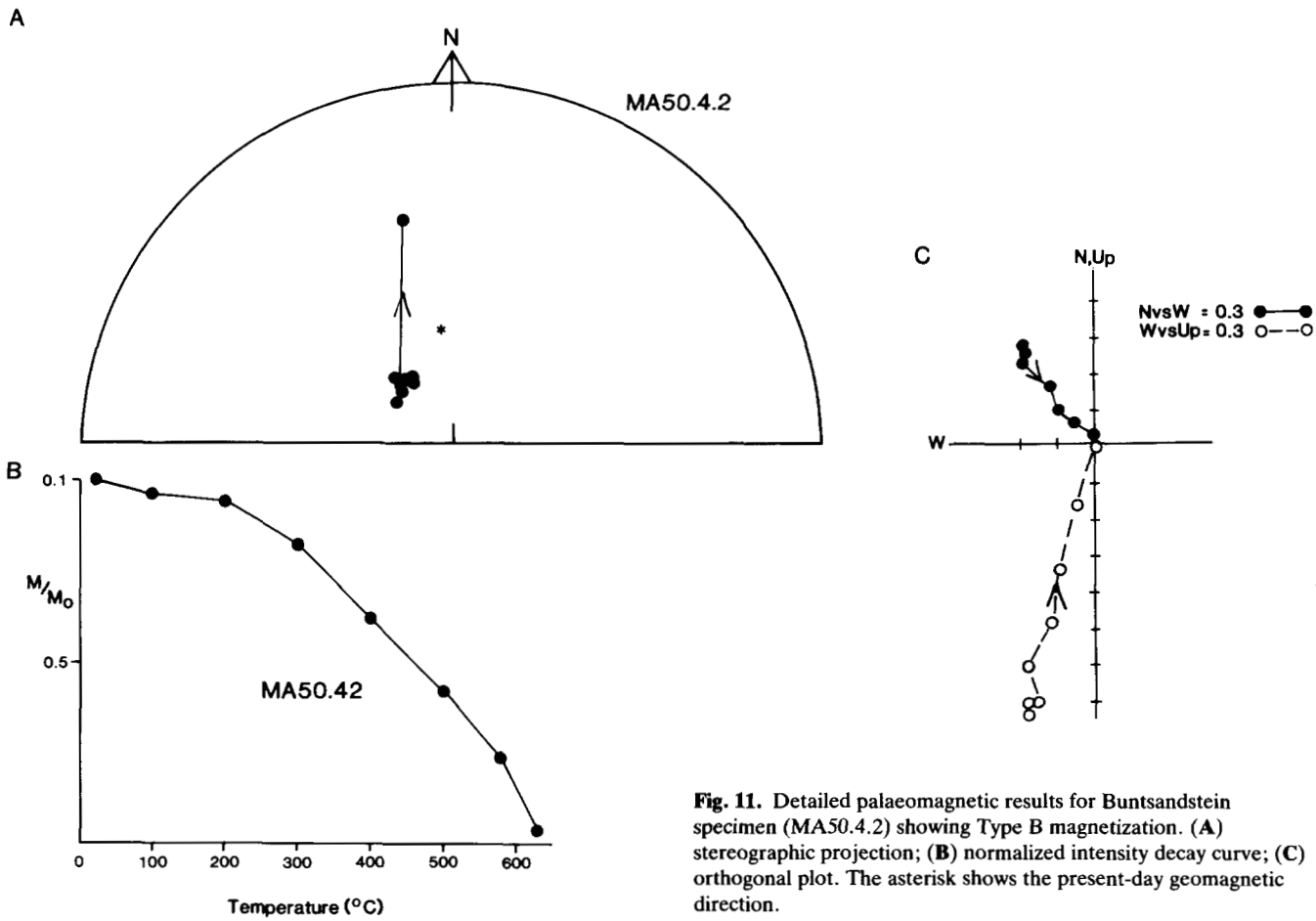


Fig. 11. Detailed palaeomagnetic results for Buntsandstein specimen (MA50.4.2) showing Type B magnetization. (**A**) stereographic projection; (**B**) normalized intensity decay curve; (**C**) orthogonal plot. The asterisk shows the present-day geomagnetic direction.

a small proportion of the original remanence is preserved in the rock.

The interpretation of magnetizations of this type is that they represent sediments which are almost totally remagnetized but which retain a vestigial record of the original remanence. In the illustrated example this original remanence is reversed; in other cases directional movement is towards the NW quadrant and in such cases the original remanence is judged to be normal. An example is shown in Fig. 11 (MA 50.4.2). In this case notice that the intensity decay is almost identical to MA 49.4.1. but the directional movement is towards the N. We believe therefore that whilst magnetizations of this type are unsuitable for precise palaeomagnetic determinations we can use them to establish the magnetic polarity of the Permo-Triassic sequence (without prior assumption that the age of remanence and depositional age are closely coincident—see later discussion).

Type C

Some 12% of the Buntsandstein have magnetization of this type. They are very similar to Type B but differ in that directional movement is very slight and there is no positive evidence of any original remanence. An example is shown in Fig. 12 (MA 74.4.1). Again the initial direction is near to the local geomagnetic direction and there is no change in direction during the thermal demagnetizion procedure. Note that at 630 °C some 40% of the original remanence remains which contrasts with the Type B magnetizations. At 660 °C there is a very sharp drop in NRM intensity with only 1% of the remanence remaining. The intensity (0.02 mA m^{-1}) is close to the detection limit of the cryogenic magnetometer and the direction, steeply positive in the NE quadrant is only little removed from the original direction. We therefore interpret this type of magnetization to be the result of complete remagnetization. The fact that the directions are closely grouped near the local geomagnetic field and exclusively of the same polarity strongly suggests that the remagnetization occurred in Holocene times during the Brunhes chron and the directions thus effectively represent the present Earth's field. More detailed discussion of the recent components will be given in the next section.

Muschelkalk Facies

The Muschelkalk facies comprises a well bedded sequence of finely crystalline dolostones more weakly magnetized than the underlying clastic facies. The average NRM intensity prior to thermal demagnetization averages (0.15 mA m^{-1}). The initial NRM directions are all steeply positive and near to the local geomagnetic field direction (as in the underlying Buntsandstein facies). During partial thermal demagnetization the NRM decays steadily and there is usually a marked blocking temperature at 400 °C.

A typical example of the thermal demagnetization of a Muschelkalk dolostone is shown in Fig. 13 (MA 101.2.2). The NRM direction remains close to the local field direction

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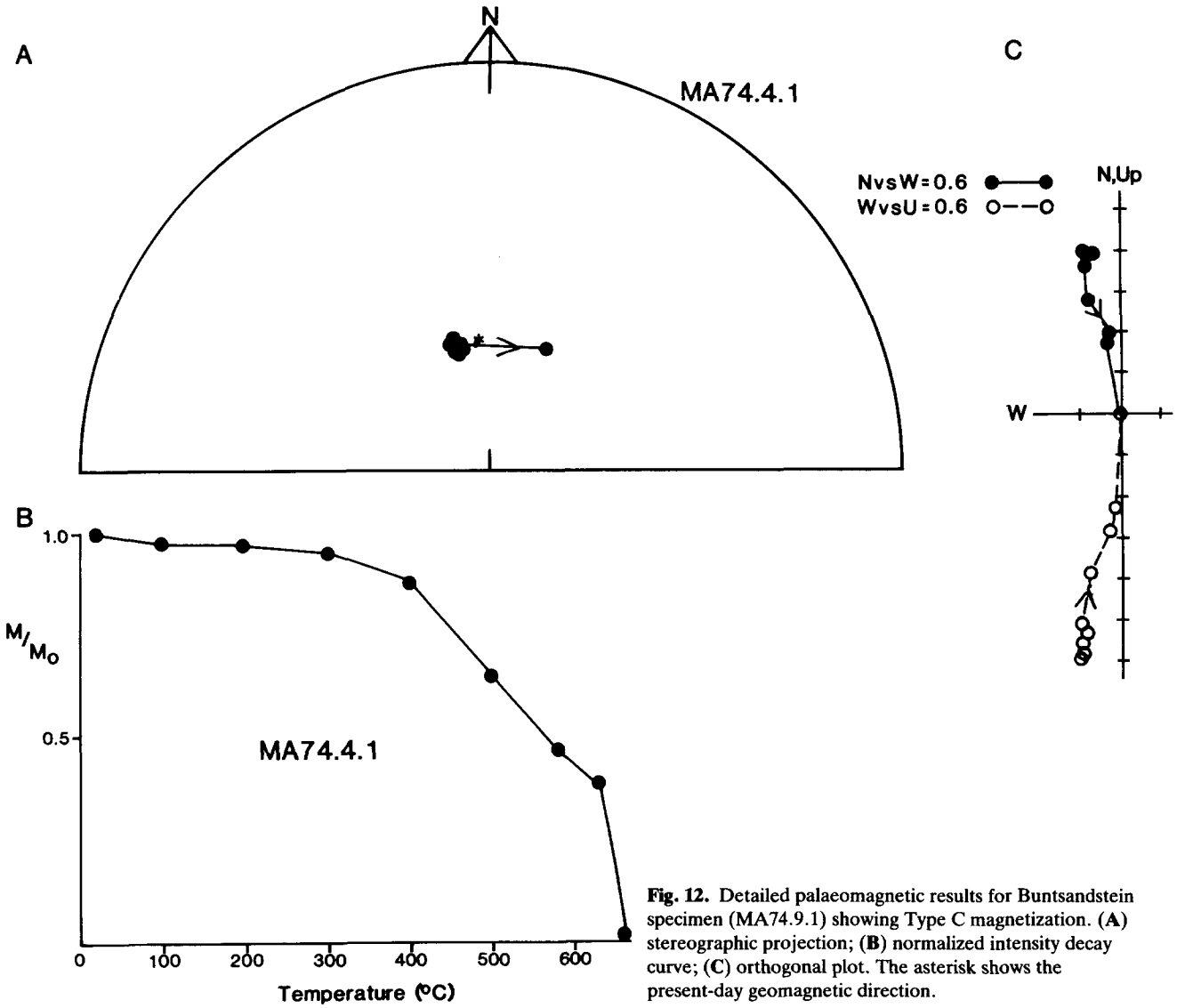


Fig. 12. Detailed palaeomagnetic results for Buntsandstein specimen (MA74.9.1) showing Type C magnetization. (**A**) stereographic projection; (**B**) normalized intensity decay curve; (**C**) orthogonal plot. The asterisk shows the present-day geomagnetic direction.

until above 400 °C when it moves towards a shallower inclination in the NW quadrant. Most of the samples behave consistently in this manner enabling the estimation of characteristic components. These components are usually isolated between 500 and 580 °C. Above 580 °C the specimens are too weak for measurement with the cryogenic magnetometer (intensity 0.01 mA m^{-1}). Fig. 14 shows the components isolated from the Muschelkalk dolostones. They are all in the NW quadrant with shallow positive inclinations with two exceptions (MA 100.1.1, MA 104.2.2). The two exceptions have components with a westerly direction and shallow negative inclination. It is noteworthy that these two specimens occur between a zone of magnetization established as reversed in the Upper Buntsandstein and the exclusively normal magnetozone of the Muschelkalk. On this basis these anomalous directions in the Muschelkalk are interpreted as intermediate or composite magnetizations in the overlap between a normal and reversed period. The fact that the NRM is so weak (0.15–0.21 mA m^{-1}) precludes any possibility of discrimination by further demagnetization methods.

Magnetic mineralogy and acquisition of remanence

In order to determine the magnetic mineralogy of the respective rock units, we have measured isothermal remanence (IRM) acquisition curves (Dunlop 1972) and studied a large number of polished sections using reflected light microscopy. The IRM acquisition curves can be classified into two groups, A and B. The former show a steady increase in IRM with field indicating that only hematite is present. The latter show a slight tendency towards saturation in the range 0.2–0.6 T and may contain small amounts of magnetite. Examples are shown in Fig. 15.

The opaque content of the Autunian deposits is small (<2%) and mainly consists of fine grained hematite and iron–titanium oxides (predominantly ilmenite) disseminated throughout the rocks. Phyllosilicates such as biotite occasionally show replacement along cleavages by small laths of hematite. Evidence for the former existence of iron sulphides (pyrite) is seen in places but hematite pseudomorphs are all that remains now. Finally, small amounts of fine grained authigenic hematite line dissolution voids in

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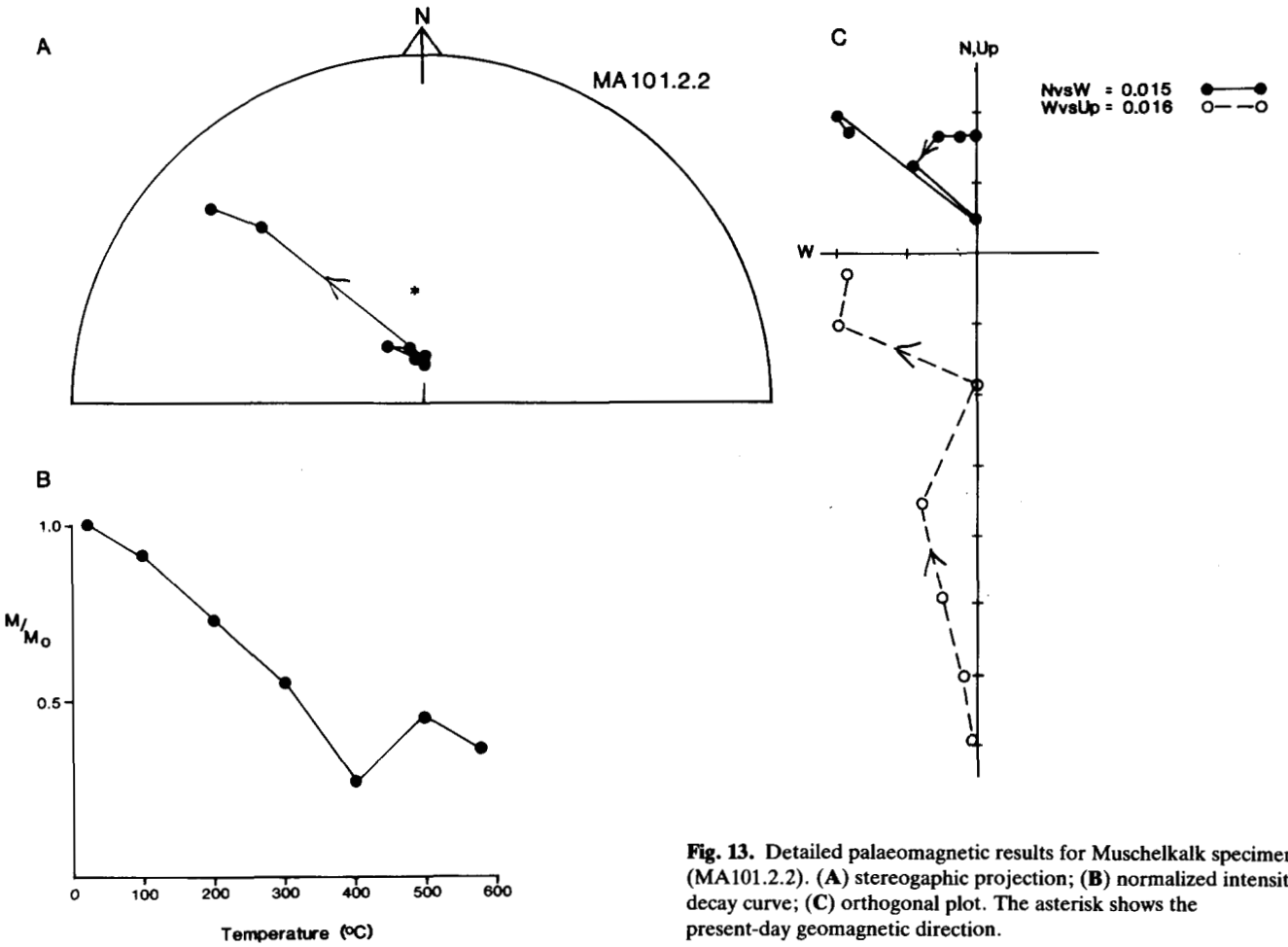


Fig. 13. Detailed palaeomagnetic results for Muschelkalk specimen (MA101.2.2). (**A**) stereogaphic projection; (**B**) normalized intensity decay curve; (**C**) orthogonal plot. The asterisk shows the present-day geomagnetic direction.

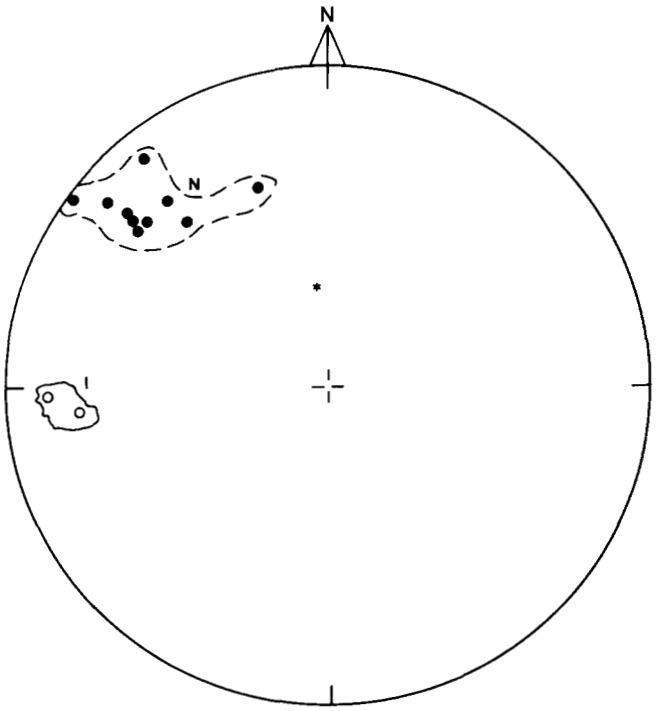


Fig. 14. Stereographic projection of components of magnetization isolated from the Muschelkalk. The asterisk shows the present-day geomagnetic direction.

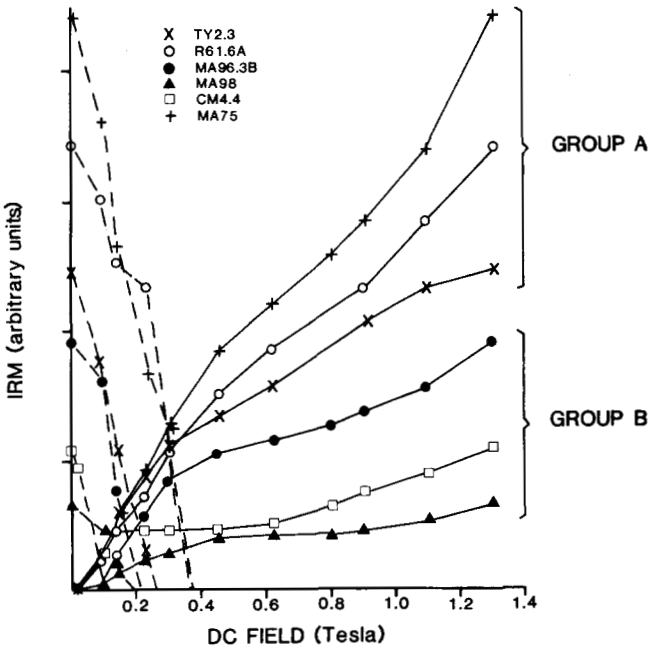


Fig. 15. Isothermal remanence (IRM) curves from representative specimens. Groups A are red sandstones which have only hematite as a magnetic phase. Group B are dark siltstones and dolostones which are believed to contain hematite plus a small amount of magnetite.

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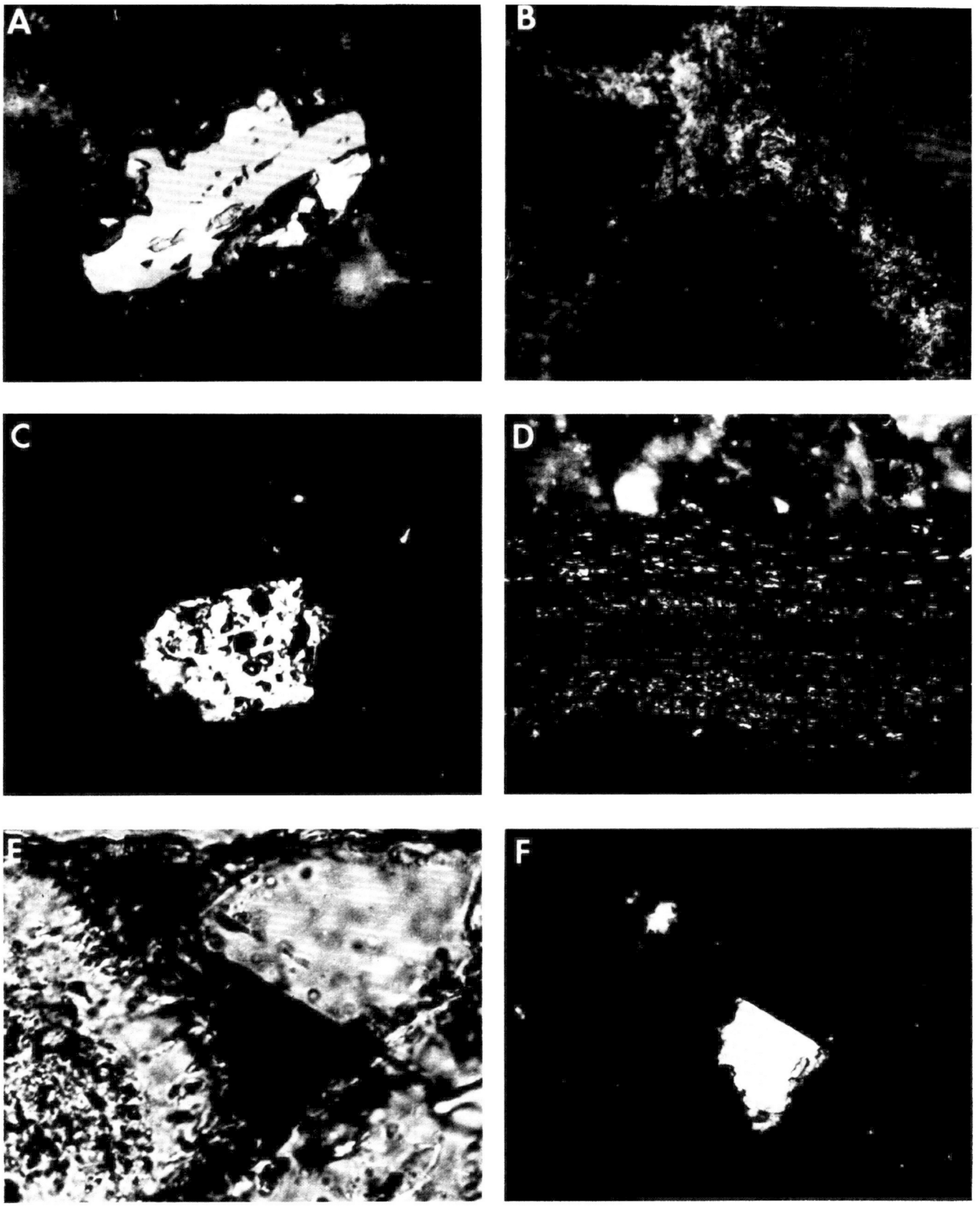


Fig. 16. Photomicrographs of detrital and authigenic iron oxides in the Buntsandstein. **(A)** Composite grain of hematite and titanium oxide with a strongly embayed appearance (reflected light, oil). **(B)** Pore-filling microcrystalline hematite and clay minerals (reflected light, oil). **(C)** Detrital hematite with pitted appearance and authigenic projections on left and right side (reflected light, oil). **(D)** Biotite partially replaced by hematite crystals aligned parallel to cleavage (reflected light, oil) **(E, F)**. The same area showing pore-filling authigenic hematite post-dating syntaxial quartz overgrowths (E, thin section in ppl, F, reflected light, oil).

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some places. IRM acquisition curves (Fig. 15) show that Autunian rocks correspond to the Group B type indicating that they may contain small amounts of magnetite. This material is believed to be responsible for the characteristic Permian remanence which was acquired early probably as a depositional or post-depositional magnetization.

In contrast to the Autunian, the Saxonian facies contain abundant iron and titanium oxides in a variety of textural phases. No martites are present but detrital hematites (specularites) are fairly common. These specularites range in size up to 80 μm and frequently show alteration having embayed margins or a skeletal habit. Detrital titanium oxides are also present, these are generally more euhedral (often tabular) then the specularite grains. Many detrital grains contain both iron and titanium oxide (Fig. 16A). Authigenic hematite is present in a number of textural forms. It occurs as discrete authigenic crystals in the matrix, as syntaxial overgrowths on detrital grains and less commonly as pore filling cement. Some rock fragments also show cementation by hematite and/or titanium oxide.

Authigenic titanium oxide is abundant in the Saxonian and in some cases may represent the daughter product of titanomagnetites, now completely altered. Biotites sometimes show replacement by hematite along cleavage traces. Much fine-grained Fe oxide is found in the Saxonian. This occurs as grain coatings and pore fillings in association with clay minerals. It is also present as a replacement of detrital hematites, eg. within centre of skeletal grains. IRM acquisition curves (e.g. TY2.3 in Fig. 15) indicate that hematite is the sole magnetic phase present.

The complexity of iron oxide textural phases and magnetic remanence is quite expected. It is clearly not easy to relate specifically which textural phase carries which magnetic component but the following scheme is suggested. Frequently, the Saxonian rocks contain three components of magnetization with significantly different blocking temperature spectra. The lowest blocking temperature component is in the present geomagnetic field direction and was presumably acquired recently.

We believe that this component (e.g. A in Fig. 7) resides in fine pigmentary hematite which may be near or just above the superparamagnetic threshold. Although this material may have formed during early diagenesis, elevated temperatures during burial would result in superparamagnetic behaviour. Remanance was locked in only after uplift to the near surface. Higher blocking temperature components are attributed to normal and reversed Permo-Triassic components of magnetization and may be associated with detrital and authigenic pore-filling oxides.

Such components are characteristically carried by coarser grained hematite but it is difficult to determine the precise textural site in which different components reside. Using MA115 (Fig. 7) as an example, the authigenic opaque hematite is generally finer than detrital grains and based on this component B could reside in authigenic phases and component C (the highest blocking temperature) could be associated with detrital grains. The characteristic magnetization was mostly acquired during deposition and early diagenetic precipitation of authigenic hematite.

The magnetic mineralogy of the Buntsandstein differs from that of the Saxonian in two important ways: (1) There are fewer detrital hematite grains; (2) There is considerably more authigenic hematite found in association with biotite micas which are abundant in the Buntsandstein.

Martites are once again absent and the few coarse detrital hematites which are present are usually very altered having a ragged, embayed and skeletal appearence. Detrital ilmenites are also present, many showing replacement by fine grained iron and titanium oxide. The most common textural phase of hematite in the Buntsandstein is microcrystalline or pigmentary hematite. In the fine grained upper part of the Buntsandstein it is abundant in shales and siltstones where it probably formed by deposition and the early diagenetic inversion of detrital iron hydroxides.

In coarser grained lithologies fine-grained hematite occurs in pore spaces in intimate association with clay minerals (Fig. 16 B). Much of this material appears to fill secondary porosity and has resulted from the dissolution of iron-bearing carbonate cements. It seems likely that this material is an important secondary magnetization carrier. Other workers have noted the relationship between secondary magnetization and alteration of carbonate cements, particularly dedolomitization (Elmore *et al.* 1985). Other authigenic phases which must carry secondary magnetizations in the Buntsandstein include small authigenic overgrowths on detrital grains (Fig. 16 C), hematite replacements of phyllosilicates (Fig. 16 D) and pore-filling hematite which clearly post-dates the precipitation of syntaxial quartz overgrowths (Fig. 16 E, F).

Discussion

Extending the established polarity time scale backwards in time through the Permian and Triassic has several potential advantages providing that problems concerning the aquisition of remanence can be adequately resolved. The most important of these is the correlation of continental sequences which lack biostratigraphical control. Although there appear to be several fairly short-lived reversals in the Lower Triassic which can be correlated over some distances (eg. Lienert & Helsley 1980) there is no independent geochronology evidence that the age of the rocks and the reversals are closely coincident and although the palaeomagnetic evidence might be used to argue for early acquisitions of the remanence (eg. Purucker *et al.* 1980) this is not consistent with the diagenetic evidence of the rocks (Larson *et al.* 1982). The problem has really been highlighted by the discussion of a postulated polarity transition record in the Chugwater Formation (Herrero-Bervera & Helsey 1983; Larson & Walker 1985; Helsey & Herrero-Bervera, 1985). Here, intermediate directions in a 1 m thick interval between N–R polarity transition are interpreted as a faithful recording of the geomagnetic field during transition (Herrero-Bervera & Helsey, 1983). This clearly requires rapid remanence acquisition for which, according to Larson & Walker (1985), there is no evidence. They argue that the same features can be generated by the superimposition of remenance components of opposite polarity with at least one of the components being of diagenetic origin. Arguments of this type will continue for a long time because of the lack of geochronological control. Red beds are largely devoid of adequate biostratigraphical control and for this reason will continue to pose particular problems in magnetostratigraphic studies. A possible way forward with this problem is to isolate the remanences which are carried by detrital and authigenic phases. However, it is clear from this study that this is not always possible.

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A very significant feature of the Buntsandstein is the relationship between the authigenic fine grained iron oxides and other diagenetic features. Reconstruction of the burial history of the area shows that the Buntsandstein is unlikely to have been buried more deeply than 3–4 km. This is consistent with clay mineral analysis of 2 μm fractions which show abundant smectite and mixed layer illite-smectite. Diagenesis took place partly under relatively shallow burial in a manner similar to that described by Walker *et al.* (1978) and Burley (1984). This included dissolution of silicate phases and precipitation of authigenic quartz, potash feldspar, smectite, kaolinite and carbonates. During the later stages of diagenesis, and especially inversion of the Permo-Triassic sedimentary basins carbonates were dissolved and a residual product of this reaction was the formation of iron oxides. Secondary porosity generation of this type is extensive in the Buntsandstein and fine grained oxides are frequently seen lining and filling secondary pores (Fig. 16B). It is these phases that are considered to carry the Holocene secondary magnetization of the Buntsandstein.

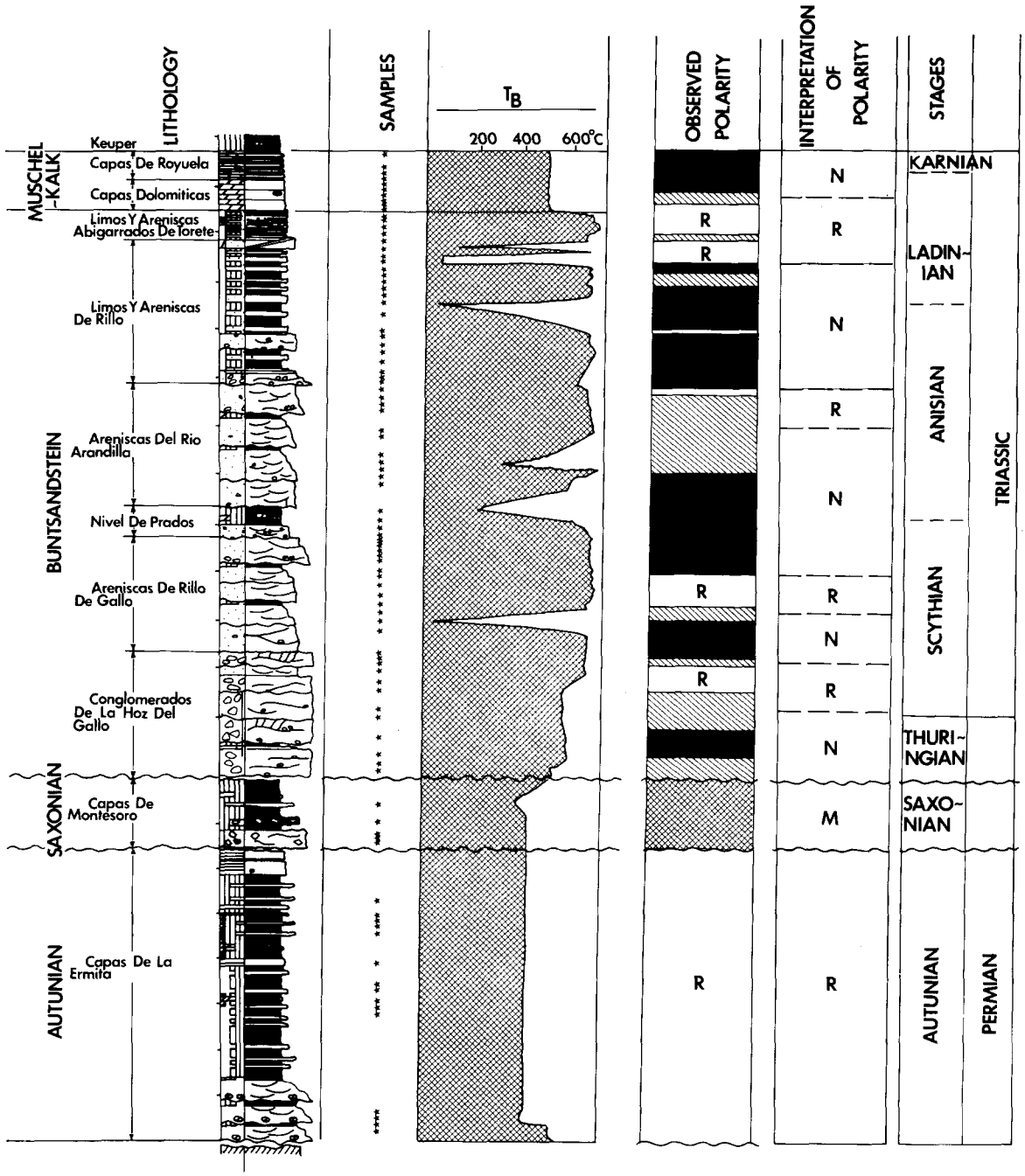


Fig. 17. Composite of lithostratigraphy, sampling horizons, blocking temperature at which secondary magnetization is removed and the interpreted polarity of the characteristic magnetization.

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The obvious implications is that secondary porosity development is very Recent and probably associated with the present ground water zone.

Another possibility is that the secondary porosity was formed when the sediments were more deeply buried and that the iron oxides produced at this time were in the superparamagnetic state and carried no permanent remanence.

The higher blocking temperature component must be associated with coarser iron oxides which are predominately detrital in origin. The direction of these components are certainly pre-Cretaceous since they are rotated relatively to stable Europe (Van der Voo 1969) and by comparison with other data from Spain are considered to be Triassic.

We have attempted to construct a magnetostratigraphy for the Molina de Aragon section based on our palaeomagnetic analysis (Fig. 17). This figure shows the sampling horizons in relation to a summary of the sedimentological variation. The minimum blocking temperature of components used for polarity determination is indicated and it should be noted that this gives an immediate visual impression of the most intensely remagnetized zones (T_b column in Fig. 17).

It should be noted that this section is based on higher blocking temperature components which we believe have survived the pervasive Holocene remagnetization seen in these rocks.

For each sample horizon an observed polarity has been determined on the basis of isolated components or directional trends as discussed in earlier sections (observed polarity column in Fig. 17). At this stage of the analysis, no implications are made regarding the absolute age of the remanence (polarity). Four categories have been used: Normal, reversed, intermediate and mixed. Sample horizons have been designated normal or reversed only when directions are clearly defined on the basis of delination and inclination. Intermediate polarity is determined with declinations and/or inclinations which lie in quadrants other than normal or reversed. No attempt has been made to discriminate intermediate components which may be single components and representative of genuine intermediate field direction (assuming they may exist in the data) and those which comprise resultant directions of combined normal and reversed components (which almost certainly do exist in the data).

The fourth category of mixed polarity is used exclusively in Fig. 17 for the rocks of the Saxonian facies. In every specimen from this unit superimposed normal and reversed components were identified. It is not clear whether this represents a series of short-lived reversals so that discrete zones cannot be identified or whether it is due to overlap between a normally magnetized zone and the underlying reversed Kiaman zone (see Fig. 17).

Using these criteria a number of stratigraphically significant features can be identified. Autunian section is completely reversed, it is considered to correspond to the Kiaman Epoch which has been widely identified in late Carboniferous and Permian rocks (Irving & Parry 1963). A period of normal and reversed zones appears to characterize the latest Permian and Scythian with two prominent normal events in the latest Scythian and Anisian. The Ladinian Stage corresponds to an important reversal and the Karnian appears exclusively of normal polarity. No correlation with other Triassic sections has been attempted primarily because of the uncertainities arising due to lack of biostratigraphical control.

Conclusions

Combined mineralogical and palaeomagnetic studies of Permo-Triassic rocks from the Iberian Cordillera, Spain enables the discrimination of components which were acquired before and after the rotation of the Iberian peninsula during Cretaceous times. Lower Permian (Autunian) rocks have an early-acquired characteristic remanence which is exclusively reversed and corresponds to the Kiaman Epoch. Even though some Autunian rocks are not strongly oxidized they contain secondary magnetization with blocking temperatures up to 400 °C suggesting late-acquisition during burial and uplift, after the rotation of the Iberian Cordillera.

Late Permian (Saxonian) red beds are characterized by complex magnetization in which normal and reversed Permo-Triassic components are usually superimposed. These were acquired during the early diagenetic alteration and authigenesis of iron–titanium oxides. As in the Autunian rocks, a late acquired component associated with burial and uplift is present.

The Buntsandstein (Late Permian-Triassic) in the Iberian Cordillera shows evidence of characteristic magnetization in a number of fine grained lithologies which reveal normal and reversed zones of magnetization. The most remarkable feature of the Buntsandstein is that much of the formation has been completely remagnetized with blocking temperatures usually over 600 °C. This secondary magnetization lies close to the present day local geomagnetic direction and is associated with pore-lining and pore-filling iron oxides. These are associated with secondary porosity generated by carbonate cement dissolution. These authigenic iron oxides cannot be used to absolutely date this diagenetic event because they were initially in the superparamagnetic stage at depth and the remanence was only blocked in by relatively recent structural inversion of the Iberian Cordillera. The results indicate that palaeomagnetic results from continental red beds may be used to date diagenesis, burial and uplift in sedimentary basins.

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